

Prospectus

Jonah Cooke

Advisor: Yuan Shi

Plasma Quantum Group

Department of Physics, University of Colorado at Boulder

April 17, 2026

Brief background to dark matter

In 1980, Vera Rubin published a paper that is credited with providing the first substantial evidence for dark matter (Rubin, Ford, and Thonnard 1980). However, no one has been able to identify the substance beyond speculation. Reasonable efforts have failed to describe the rotation curves Rubin first noticed, which implies that a new direction needs to be tried. Modern plasma physics gives insight on what to try.

Background to research

When a photon travels through a plasma medium, its energy is conserved. This means that the plasma frequency and wave number of the photon must change inversely, where the plasma frequency is dependent on the density of the plasma. This implies that as a photon travels through a plasma with varying density, if the plasma density goes up, the wave number of the photon must go down. Using Snell's law, the photon's angle must increase away from the path of travel, meaning that through a plasma gradient it diverges from its initial path. This behavior is based on the equation:

$$\omega^2 = \omega_p^2 + k^2 \quad (\text{natural units assumed}),$$

which is similar in form to Einstein's equation:

$$E^2 = m^2 + p^2.$$

This implies that we may be able to treat mass as a variable and obtain a similar effect. However, the treatment is more complex, as the momentum is inherently mass dependent and the opposite dependence is necessary for the observed galactic curves.

My planned research

A detailed treatment of the one-dimensional case can be found in (Shi 2021). However, the interaction of orbital bodies is two-dimensional, and work needs

to be done to extend the research to two dimensions. This can be done by modifying the Schwarzschild action to encompass a non-constant mass distribution and extremizing.

The one-dimensional case gave a differential equation for the mass distribution, and using a known trajectory, I was able to find a mass distribution that agreed with that trajectory. The two-dimensional case is inherently more difficult. Although an analytic solution to the Schwarzschild geodesic equation exists (Hackmann and Lämmerzahl 2008), it is much more common to use numerical solutions because of the complexity of the Schwarzschild geodesics.

In addition to having the potential to describe the observed behavior of sparse matter in galaxies, this approach offers great flexibility. Despite its complexity is likely on par Schwarzschild geodesic equation, it will reduce the geodesic equation to an ODE in mass. If an analytic mass distribution exists, this will allow for a general comparison to observed data and theoretical results. Which means that if uniqueness can be proven for the result, it will either be a solution to all observed trajectories or to only one observed trajectory depending on its alignment to observational data. The other options are it doesn't have an analytic solution, or it isn't unique. Both will give insight in whether or not spatial mass distributions are a valid description of the observed motion.

Timeline:

November-December: Finished interview problem

January-February: Solved a modified version of interview problem for mass distribution based on trajectories found in interview problem. Proved uniqueness.

March-April: Began working on preliminary math to start working on the orbital case. I found a non-relativistic form of the modified potential.

May: I aim to have a form of the mass distribution (not necessarily analytic) and a trajectory to match it to.

June-July: I aim to have an idea of whether an analytic form of the mass distribution is possible, and have a body of papers containing observational data to compare the model to. If the solution I find is analytic, I plan to have uniqueness proved or disproved at this point.

August: I hope to have the bulk of my thesis writing done by the end of August.

References

Cited material

- Hackmann, Eva and Claus Lämmerzahl (May 2, 2008). “Complete Analytic Solution of the Geodesic Equation in Schwarzschild–(Anti-)de Sitter Space-times”. In: *Physical Review Letters* 100.17, p. 171101. DOI: 10.1103/PhysRevLett.100.171101. URL: <https://link.aps.org/doi/10.1103/PhysRevLett.100.171101> (visited on 04/15/2026).
- Rubin, V. C., W. K. Ford Jr., and N. Thonnard (June 1, 1980). “Rotational Properties of 21 SC Galaxies with a Large Range of Luminosities and Radii, from NGC 4605 (R=4kpc) to UGC 2885 (R=122kpc).” In: *The Astrophysical Journal* 238, pp. 471–487. ISSN: 0004-637X. DOI: 10.1086/158003. URL: <https://ui.adsabs.harvard.edu/abs/1980ApJ...238..471R> (visited on 04/15/2026).
- Shi, Yuan (Sept. 3, 2021). *Force, Metric, or Mass: Disambiguating Causes of Uniform Gravity*. DOI: 10.48550/arXiv.1908.02159. arXiv: 1908.02159 [gr-qc]. URL: <http://arxiv.org/abs/1908.02159> (visited on 04/14/2026). Pre-published.

Supplementary material

- Goldston, R.J. and P.H. Rutherford (2020). *Introduction to Plasma Physics*. CRC Press. ISBN: 978-1-000-29175-9. URL: <https://books.google.com/books?id=OrGNEQAAQBAJ>.
- Misner, C.W. et al. (2017). *Gravitation*. Princeton University Press. ISBN: 978-0-691-17779-3. URL: <https://books.google.com/books?id=SyQzDwAAQBAJ>.
- Rubin, V. C., W. K. Ford Jr., and N. Thonnard (Nov. 1, 1978). “Extended Rotation Curves of High-Luminosity Spiral Galaxies. IV. Systematic Dynamical Properties, Sa $\rightarrow$ Sc.” In: *The Astrophysical Journal* 225, pp. L107–L111. ISSN: 0004-637X. DOI: 10.1086/182804. URL: <https://ui.adsabs.harvard.edu/abs/1978ApJ...225L.107R> (visited on 02/09/2026).
- suvojit1999 (Feb. 11, 2026). *Suvojit1999/Simulation-of-perihelion-precession*. URL: <https://github.com/suvojit1999/Simulation-of-perihelion-precession> (visited on 04/17/2026).
- Taylor, J.R. (2004). *Classical Mechanics*. MIT Press. ISBN: 978-1-891389-22-1. URL: <https://books.google.com/books?id=t41pEQAAQBAJ>.